KVM/QEMU Hypervisor Lab

Day 1 Lab Journal — Build Log & Incident Record

Date: June 22, 2026 | Engineer: Demi | Host OS: Ubuntu 24.04.3 LTS

Placement: Navy QUORRA — SIWES Industrial Training

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1. Objective

Design and deploy a production-grade KVM/QEMU hypervisor on Ubuntu 24.04.3 LTS with an optimised three-disk storage layout, hosting a Windows Server 2022 Active Directory domain controller as the core identity server domain

2. Hardware & Disk Layout

2.1 Target Storage Design

Disk	Role	Contents
nvme0n1 (238.5GB SSD)	System + Active VMs	Ubuntu host (/), /boot/efi, /boot, /var/lib/libvirt/images
sda (465.6GB HDD1)	Data + Files	/mnt/data — ISOs, file shares, backups
sdb (465.6GB HDD2)	Backup Mirror	/mnt/backup — VM snapshots, DB dumps, AD backup

3. Phase 0 — Disk Verification

3.1 What We Did

Before touching any storage, we ran lsblk to map all physical disks and confirm the Ubuntu host disk, HDD1, and HDD2.

Command Run:

```
lsblk -o NAME,SIZE,TYPE,MOUNTPOINT,FSTYPE
```

Error Encountered — lsblk --help output instead of disk list

The first attempt accidentally triggered the lsblk help page (showing all column definitions) instead of the disk table. This was because the command was mistyped or run with a --help flag accidentally.

- Resolution: Re-ran the correct command with explicit column flags

- Lesson: lsblk with no flags shows a basic tree; always specify -o for a clean targeted output

Confirmed Disk Layout:

Device	Size	Type	Mountpoint	Filesystem
nvme0n1	238.5G	disk	-	-
nvme0n1p1	1G	part	/boot/efi	vfat
nvme0n1p2	2G	part	/boot	ext4
nvme0n1p3	235.4G	part	/	LVM/ext4
sda	465.6G	disk	-	NTFS (old)
sdb	465.6G	disk	-	NTFS (old)
sr0	1024M	rom	-	-

Note: sda and sdb both had existing NTFS partitions from prior Windows use. Both were confirmed safe to wipe before proceeding.

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4. Phase 0.5 — KVM/QEMU & libvirt Installation

4.1 CPU Virtualisation Check

```
egrep -c '(vmx|svm)' /proc/cpuinfo
```

Returned a non-zero value, confirming Intel VT-x/AMD-V support in hardware.

4.2 Network Failure — Package Installation Blocked

Attempted to install cpu-checker and the KVM stack. apt failed to fetch packages with the following error:

```
Temporary failure resolving 'archive.ubuntu.com'
```

Root cause: The hypervisor had no network connectivity. Diagnosis steps run:

- ping 8.8.8.8 — timed out (no carrier, not a DNS issue)
- ip addr show enp2s0 — showed NO-CARRIER state
- Interface was UP but no physical link detected

Resolution:

- Physical ethernet cable was plugged into the machine
- Ran: sudo nmcli device connect enp2s0
- ping 8.8.8.8 succeeded after cable connection

Lesson: Always verify physical network connectivity before attempting package installation on a headless or newly configured hypervisor.

4.3 Time Sync Error — Release File Not Valid Yet

After fixing the network, apt update returned:

```
Release file for noble-security/InRelease is not valid yet
```

Root cause: System clock was set to a date in the past, causing Ubuntu's package release files (signed with future timestamps) to be rejected.

Resolution:

```
sudo timedatectl set-ntp true
```

Clock synced via NTP. apt update ran cleanly after.

4.4 Package Install — Command Formatting Error

Attempted multi-line install using backslash line continuation in the terminal. All packages were concatenated into one invalid string:

```
E: Unable to locate package qemu-kvmqemu-utilslibvirt-dae...
```

Root cause: The backslash continuation did not parse correctly in the terminal session, merging all package names into one token.

Resolution:

Installed packages on a single line with space separation:

```
sudo apt install -y qemu-kvm qemu-utils libvirt-daemon-system
libvirt-clients bridge-utils virtinst virt-manager ovmf libguestfs-tools
```

All packages installed successfully.

4.5 User Group Configuration

```
sudo usermod -aG libvirt,kvm bravesoldier
```

Initial groups check showed libvirt was added but kvm was missing. Ran a second command to add kvm explicitly:

```
sudo usermod -aG kvm bravesoldier
```

Final groups output confirmed: bravesoldier adm cdrom sudo dip plugdev lxd libvirt kvm

4.6 Verification Results

Check	Result	Status
kvm-ok	KVM acceleration can be used	**SUCCESS**
libvirtd service	active (running)	**SUCCESS**
virsh version	libvirt 10.0.0 / QEMU 8.2.2	**SUCCESS**
virsh net-list	Empty (no default network yet)	**NOTE**
/dev/kvm	Exists and accessible	**SUCCESS**

5. Phase 1 — Storage Setup

5.1 Format HDD1 (sda) as Data Drive

```
sudo wipefs -a /dev/sda
sudo mkfs.ext4 -L DATA /dev/sda
```

Note: A typo on the wipefs command initially targeted /dev/sub which does not exist. Corrected immediately and re-run against /dev/sda successfully.

5.2 Format HDD2 (sdb) as Backup Drive

```
sudo wipefs -a /dev/sdb
sudo mkfs.ext4 -L BACKUP /dev/sdb
```

sdb had an existing DOS partition table. mkfs prompted for confirmation — answered Y to proceed.

5.3 Mount Points & Directory Structure

```
sudo mkdir -p /mnt/data /mnt/backup
sudo mount /dev/sda /mnt/data
sudo mount /dev/sdb /mnt/backup
```

Directories created on HDD1:

- /mnt/data/isos — Windows Server ISO storage
- /mnt/data/shares — File shares
- /mnt/data/backups — General backups

Directories created on HDD2:

- /mnt/backup/vm-snapshots — QEMU VM snapshots
- /mnt/backup/db-dumps — Database dumps
- /mnt/backup/ad-backup — Active Directory system state backups

5.4 Persistent Mounts via /etc/fstab

Retrieved UUIDs with blkid and added to /etc/fstab:

```
UUID=9a1e5c8c-fcb9-4c46-9f1a-9d948a1e56de  /mnt/data    ext4
defaults,nofail  0  2
UUID=3cc319ee-c6c2-4b4a-b608-47ab038b029c  /mnt/backup  ext4
defaults,nofail  0  2
```

Verified with sudo mount -a (no errors) and df -h:

```
/dev/sda  458G  40K  435G  1%  /mnt/data
/dev/sdb  458G  28K  435G  1%  /mnt/backup
```

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6. Phase 2 — libvirt Storage Pool

6.1 Storage Pool Creation

```
sudo mkdir -p /var/lib/libvirt/images
```

Defined a libvirt storage pool pointing to the SSD-backed images directory.

6.2 Permissions Error — Pool Inactive

Multiple attempts to start the pool failed with:

```
error: cannot open directory '/var/lib/libvirt/images': Permission denied
```

Root cause: The /var/lib/libvirt/images directory had incorrect ownership. libvirt's QEMU process runs as the libvirt-qemu user in the kvm group, but the directory was owned by root:root with restrictive permissions.

Troubleshooting Steps Taken:

- Attempted chown libvirt-qemu:libvirt — chown succeeded but pool still failed
- Attempted chown libvirt:libvirt — invalid user error (libvirt group exists but not as a user)
- Ran ls -la /var/lib/libvirt/ to inspect actual ownership — confirmed libvirt-qemu:kvm on the images directory
- Ran pool-destroy and pool-undefine to reset the broken pool state

Resolution:

```
sudo chown libvirt-qemu:kvm /var/lib/libvirt/images
sudo chmod 755 /var/lib/libvirt/images
sudo virsh pool-define-as default dir --target /var/lib/libvirt/images
sudo virsh pool-start default
sudo virsh pool-autostart default
```

6.3 Final Pool Status

Pool	State	Autostart
default	active	**SUCCESS**

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7. Errors & Resolutions Summary

Error	Root Cause	Resolution
lsblk showing help page	Command mistyped — triggered --help output	Re-ran with correct flags: lsblk -o NAME,SIZE,TYPE,MOUNTPOINT,FSTYPE
apt: temporary failure resolving archive.ubuntu.com	No physical network connection (NO-CARRIER)	Plugged ethernet cable; interface came up automatically
Release file not valid yet (noble-security)	System clock behind — NTP not synced	sudo timedatectl set-ntp true

Unable to locate package qemu-kvmqemu-utils...	Backslash line continuation concatenated all package names	Installed all packages on one line with spaces
kvm group missing from user groups	usermod only added libvirt group initially	sudo usermod -aG kvm bravesoldier ran separately
Pool Permission Denied	Wrong ownership on /var/lib/libvirt/images	sudo chown libvirt-qemu:kvm + chmod 755
wipefs: /dev/sub not found	Typo in device name (sub instead of sdb)	Re-ran correctly against /dev/sdb

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8. End of Day 1 Status

Milestone	Status
Ubuntu 24.04.3 host verified and running	✅ **Complete**
KVM hardware acceleration confirmed	✅ **Complete**
QEMU 8.2.2 + libvirt 10.0.0 installed	✅ **Complete**
bravesoldier added to libvirt + kvm groups	✅ **Complete**
HDD1 (sda) formatted ext4, mounted at /mnt/data	✅ **Complete**
HDD2 (sdb) formatted ext4, mounted at /mnt/backup	✅ **Complete**
fstab entries added for persistent mounts	✅ **Complete**
libvirt default storage pool active on SSD	✅ **Complete**
Lab network (labnet 192.168.10.0/24) creation	⏳ **Day 2**
Windows Server 2022 ISO download	⏳ **Day 2**
DC01 VM creation and AD DS installation	⏳ **Day 2**

Day 2 — Next Steps

- Phase 3: Create labnet virtual network (NAT mode)
- Download Windows Server 2022 Evaluation ISO to /mnt/data/isos
- Create 80GB qcow2 disk image for DC01
- Deploy DC01 VM via virt-install
- Install AD DS, DNS, DHCP roles and promote DC01 as domain controller
- Take baseline snapshot of DC01 and copy to /mnt/backup/vm-snapshots

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